Advanced SCADA Concepts

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The typical utility system planner faces a wide variety of concepts and applications to consider when procuring a supervisory control and data acquisition (SCADA) system. The system's performance requirements are influenced by an arena of rapidly advancing technologies and a highly competitive business environment. Unlike other elements of the electric utility industry, these changes are not evolving specifically for the SCADA industry; they are being driven by other technology forums, with a profound impact on the SCADA system of the future. This article explores some of these concepts and suggests technology and business issues to consider when planning for the SCADA system of tomorrow.

A fully integrated substation system could be composed of a series of microprocessor relays interconnected to a substation controller by a high-speed communications link

Driving Forces

Very little change occurred in SCADA system concepts during the first 30 years of the industry. Specialized, dedicated computers gathered information and sent control commands over 1200-baud communications lines to remote terminal units (RTUs). The RTUs were nothing more than the name implies, remote terminals of a computer system with no local intelligence or function beyond serving the master station. The systems had little or no interaction with the rest of the utility enterprise. Consequently, SCADA system design and procurement were oblivious to other aspects of utility engineering and operation, both in the substation and at the control center. Duplication of effort was rampant, with the RTU often monitoring points that were monitored by two or three other subsystems.

The increasingly competitive business environment of utilities is causing a re-examination of SCADA as a part of the process of utility operations, not as a process unto itself. This re-examination reflects a number of business and technology concepts that are the reality of the utility industry:

- Utility operations can no longer afford the luxury of dedicated, stand-alone systems. Hardware and software of the modern SCADA system must be incorporated into the enterprise-wide, management information systems strategy to maximize the benefit to the utility. In addition, specialized efforts to install, operate, and maintain a SCADA system must be reduced or eliminated as part of the overall corporate competitive posturing.
- Substation operations are becoming automated through the use of powerful process computers and intelligent devices. These devices can interconnect to produce integrated data collection and control of the substation process for local operation as well as for SCADA. The result is an integrated protection, control, and revenue management system for electrical transmission, distribution, and delivery. These systems also have enough computing capability to off-load processes that were previously performed by the master stations. Certain of these processes can be performed more efficiently and with greater accuracy at the substation.
- Communications technology developments continue to advance and drive the SCADA industry into new directions. It is now cost feasible to have T-1 service (1.44 megabits/second) communications to every transmission substation in a utility system. This facilitates the transfer of large blocks of data, and frees the SCADA architecture and functionality from the archaic 1200-baud limitation of past communications systems. Because of these communication developments, the substation SCADA system will assume a larger roll in the utility enterprise. The advent of fiberoptics, direct satellite broadcast, and third-party digital service providers has made high data rate transmission economical.

SCADA: The Virtual System

The SCADA system of tomorrow will incorporate enter-

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prise-wide integration. The control center and its dedicated hardware and software resources will cease to exist. This will parallel the dissolution of the separate engineering, accounting, manufacturing, and word processing systems that were put in place during the 1970s and 1980s. In those companies that postured themselves to survive, the dedicated islands of automation gave way to the corporate management information system.

Similarly, in advanced systems, SCADA will be a function performed by workstation-based applications, interconnected through a wide area network (WAN) to create a virtual system (see Figure 1). This arrangement will provide the SCADA applications access to a host of other applications, e.g., substation controllers (defined later in this article), automated mapping/facility management system, trouble call analysis, crew dispatching, and demand-side load management. The WAN will also provide the traditional link between the utility's energy management system and SCADA processors.

The applications will be interconnected through the use of commercially available and independently supported formats, such as databases, graphical displays, and communications protocols. This system and interconnection will provide immediate benefits to the electric power utility seeking to meet the competitive pressures of the business environment.

The workstation-based applications will also provide for flexible expansion and economic system reconfiguration. Advanced applications modification, redundant processing, and point count expansion will be performed without disturbance to the installed operating system.

Systems based on substation controllers and LAN/WAN facilities provide tools to generate positive, quantifiable benefits to a utility's construction and operation

At the heart of this integrated system is the concept of a distributed database. Unlike the centralized database of most existing SCADA systems, the advanced SCADA system database will exist in dynamic pieces that are distributed throughout the network. Modifications to any of the interconnected elements will be immediately available to all users, including the SCADA system.

The virtual system also provides a fault-tolerant disaster recovery capability not possible with past SCADA systems/control centers. Since the SCADA system is simply a set of workstations with applications, a redundant system can be established wherever and whenever the utility chooses.

SCADA will have to become a more involved partner in the process of economic delivery and maintained quality of service to the end user. This is widely recognized. In most systems today, SCADA and the EMS operate only on the transmission and generation sides of the system. However, industry directions and competitive forces are driving utilities to seek solutions which involve the entire electric production and consumption cycle. In the system of the future, economic dispatch algorithms will include demand-side management and voltage control/reduction solutions. The virtual system will have the data and telemetry available that are necessary to enact these solutions. At the same time, the desired balance of economics, customer satisfaction, and regulatory mandates must be maintained.

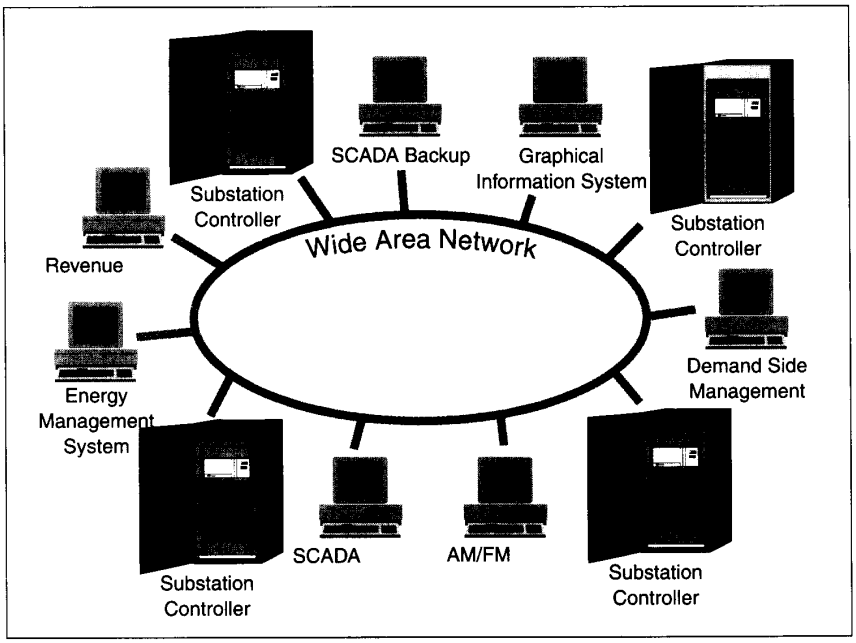


Figure 1. SCADA: the virtual system

Substation Controllers

Traditionally, the RTU has been a physical device, installed solely for the purpose of providing station telemetry and control to the master station. Equipment concepts of today and tomorrow challenge this definition. The

device acting as the RTU can also provide other primary functions such as system protection, local operation, graphical user interface (GUI), and data gathering/concentration from other subsystems.

Today's technical environment has proven that the RTU, as currently defined, may be a thing of the past. In its place is the substation controller, a class of devices that performs multiple substation control, protection, and operation functions. In conjunction with these functions, the substation controller also develops and processes data required by the SCADA master, and it processes control commands and messages received from the SCADA master.

The substation controller is a class of devices that performs multiple substation control, protection, and operation functions

Substation controllers provide advances to SCADA systems in two important ways.

- As a substation device, the substation controller is capable of gathering and organizing data from a variety of different devices. These provide operational information that has been either too costly or technically infeasible to gather with the traditional SCADA system arrangement of master and remote.
- As a powerful processing engine, the substation controller runs algorithms and derives data that has been historically developed in the master station. This data is more effectively developed by the substation controller and is available to personnel at the substation. This not only makes for a more fault-tolerant and processor-efficient system, but also makes data available to a wider range of users (e.g., substation personnel, field operations personnel, planners).

A part of the larger scheme of utility communications, the substation controller will provide a gateway function to process and transmit data from the substation to the WAN. The SCADA system, as well as the other users, will provide the necessary filtering to extract data of interest and importance to various software applications.

Substation Design and Operation

In the substation design of the future, the substation controller will play an important role in posturing the utility for a more competitive environment. The substation controller can greatly reduce capital investment in the substation by making data available to many users through functions rather than through hardware.

Substations often have a number of subsystems that monitor the same data points. Traditionally, these systems have had different users within the utility organization, and each user chose to purchase and install their own equipment. Competitive pressures on the utility will no longer permit this duplication of effort.

Integrated Hardware and Software. The substation controller is a computer system designed to operate in a substation environment. It is composed of hardware modules and software applications, which generally fall into one of three functional elements (see Figure 2):

- Data Processing Applications (DPAs): These software applications provide various users access to the data of the substation controller. They are used to provide instructions and programming to the substation controller, gather data from the substation controller, and perform sequential functions within the substation controller.

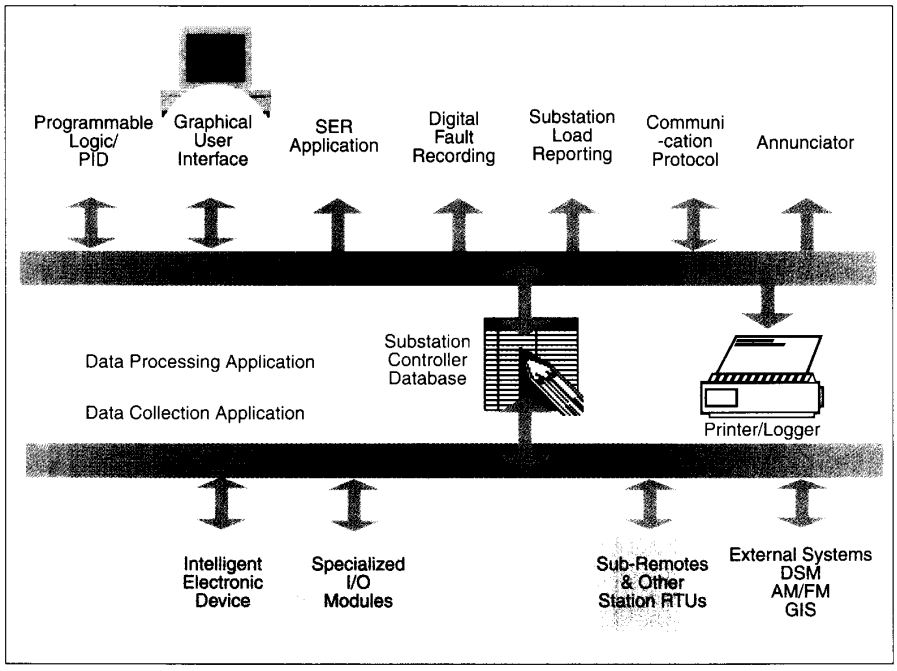


Figure 2. Substation controller

- Data Collection Applications (DCAs): These software applications provide the access to other systems and components that contain data elements necessary for the substation controller to perform its functions.
- Central Database: The key to the substation controller is its database functionality. In this facility, all data resides in a single location, whether from a DPA, DCA, or derived from the substation controller itself. This database acts as a file server and provides data to the various data concentrator and data processing applications.

By eliminating the "ownership" of data that was typical in older RTU designs, the substation controller provides the mechanisms to truly integrate substation information. Additionally, the utility is free to develop its own DPA and DCA applications as those needs arise.

It should be noted that the substation controller is not a certain piece of equipment, but actually a system. This system can be composed of many different types of hardware and software components and may not even reside in a single location. In this regard, the RTU may exist, not as a device unto itself, but as a software application within the substation controller system. In time, the SCADA DPA may eventually be completely remote from the substation. Substation controllers will make all data available on WANs. A SCADA processor located remotely will be left with the task of extracting the information required for its operation.

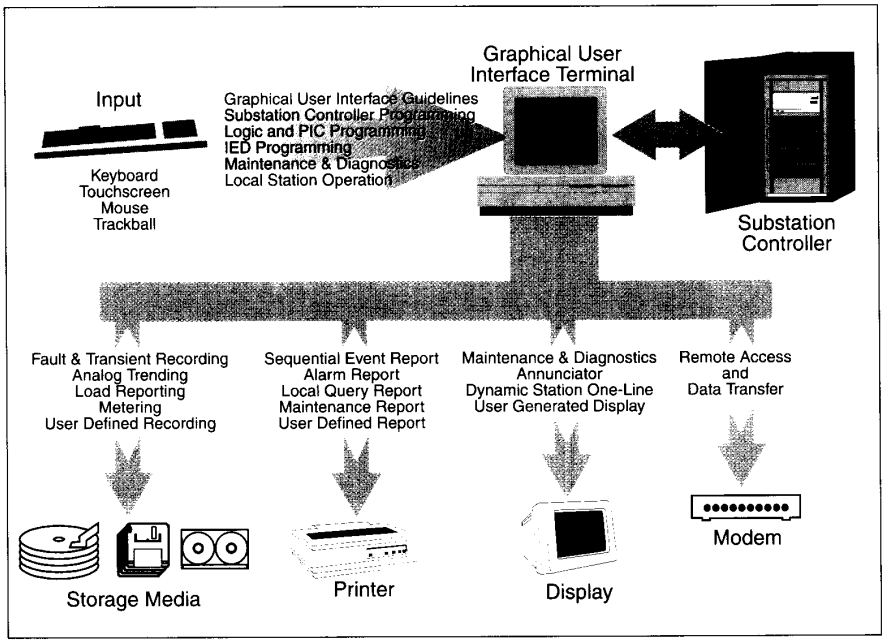


Figure 3. Substation information system

Through a series of application software programs, data from the substation controller can be extracted in numerous formats and displayed on a local CRT/Plasma display using off-the-shelf personal computer (PC) technology. These application programs will, in turn, eliminate the need for separate subsystems. The most cost beneficial programs would provide a new user interface for substation personnel, an integrated annunciator system, an integrated sequence of events recorder, an integrated digital fault and transient recorder, and programmable logic capabilities.

- GUI Application: A GUI application will present an operator with one-line diagrams and tabular displays, much like a SCADA master station. Through the GUI application, an operator can monitor and control the substation. The operator will be able to query the central database for all operating parameters of the substation and be able to send control commands to substation devices. The GUI will eliminate the need for control boards and mimic panels, meter panels, and annunciator panels. The GUI will also have the same look and feel to the substation operators, regardless of substation design, layout, and equipment. With a substation interface that is identical in each substation, training and maintenance efforts for substation operators and technicians will be greatly reduced and simplified.
- SER Application: Through a sequential event recorder (SER) DPA, the operator will be able to retrieve a sequential event file for the substation. This file can be displayed on the GUI panel, or the application can print the file to a local printer.
- DFR Application: Through a digital fault/transient recorder (DFR) DPA, the operator will be able to recover files of AC analog processed data. The data can be saved on magnetic tape or sent to fault analysis personnel over a WAN.
- Programmable Logic Application: Through a programmable logic DPA, substation data can be evaluated and decisions made. The decisions would be based on logical comparison statements and PID control.

System Benefits. The substation controller will result in a significant benefit to the utility by the elimi-

nation of separate stand-alone systems. Initial capital expenditure will be minimized, training on multiple systems will be eliminated, and demand for substation resources (panel space, dc power, real estate) will be reduced. However, the greatest savings to the utility may be achieved in the construction/installation of the substation. In traditional substation design, each separate subsystem must have independent wiring of input/output devices. The elimination of dedicated subsystems greatly reduces the installation of equipment.

In the substation controller, it is only necessary to wire the status contact once to an I/O module. The information is then made available to all of the substation applications.

Substation operations are becoming automated through the use of powerful process computers and intelligent devices

Decentralized I/O Processing. Decentralized I/O processing is accomplished through distributed nodes or processors located throughout the substation, wherever it is most convenient. The nodes communicate over a local area network (LAN) to the substation controller, which compiles the data into the central database.

RTUs with decentralized I/O processing and LAN architecture offer the engineer a greater freedom to employ the optimum physical configuration of the RTU based on the particular requirements of the substation.

Substation Protection Interface

Efforts to incorporate protection functions in a substation controller have been made in many substations. Some secondary functions, such as transfer tripping, synchrocheck protection, and reclosing have been successfully performed in an integrated substation controller platform. However, due the critical nature and specialized functionality of protection, the primary function of relaying (sensing a fault and initiating action to clear the fault) still belongs in the domain of the dedicated substation relay. In this regard, the industry approach has not changed for decades. What has changed is the nature of the relays. Newer designs in relays have abandoned the electromechanical approach in favor of microprocessor-based devices that are small-scale, specialized substation controllers. For the circuit being protected, the microprocessor relay performs its primary function, protection. In addition, it can easily acquire and develop steady-state analog values and status data and provide data for sequential events reports, DFR waveform, and breaker maintenance reports.

The advanced substation controller can access this information from the relays via a LAN. This arrangement then allows the substation controller to incorporate the relay-developed information into its central database, and serve as a gateway to the utility WAN. In effect, the relay is used as an input/output module for the substation controller and further eliminates the need for transducers and redundant wiring. Conceivably, a fully integrated substation system could be composed of a series of microprocessor relays interconnected to a substation controller by a high speed communications link.

Substation Information System

As more intelligence and interconnections are placed in the substation controller, the wealth of data (both acquired and derived) will increase. Users, beyond the SCADA system, can greatly benefit from access to this information. In the traditional SCADA system, awkward file transfers have been used to deliver SCADA-derived information for planning, substation maintenance and operation, and metering. This approach was further complicated by labor-intensive database manipulations at the SCADA master to accommodate the changing requirements of other users.

The substation controller will permit field programming and manipulation of the data collected for non-SCADA users, independent of the SCADA function. It is a relatively trivial task for the Substation Information System, depicted in Figure 3, to develop and store, print, display, and/or communicate data. Examples of specific data that can be developed include outage reports, peak loading reports, feeder demand periods, transducer calibration/accuracy errors, number of breaker operations, transformer loading reports, and I^2t fault reports. These examples and those shown in Figure 3 demonstrate how qualified users can obtain pertinent information without placing a burden on the SCADA master station system or its personnel.

Ultimately, the substation controller will be viewed as a generic data collection and communication processing unit, which provides many users in the "outside world" with different aspects of the substation process. The substation controller will be the central manager of the data, and present it to the communications network for any and all interested users in the enterprise-wide utility network.

Positive Cost/Benefit Ratio

Current economic realities mandate that SCADA must stand the scrutiny of cost/benefit analysis. As with every other major expense in utility operations today, SCADA system procurement and operation are being closely examined. The principle criteria of these examinations is "How does this expense improve the posture of the company in the growing competitive environment?".

Today, SCADA system planners will be challenged to develop a business case for the procurement of a new system, or the upgrade of an existing system. If the business case does not provide a positive, quantified cost/benefit ratio, the procurement or upgrade is unlikely to be approved.

The advanced SCADA system of the future, and the SCADA system planner must meet this challenge. The planner, in the role of a business manager, can use the features and capabilities of the advanced SCADA system as an economic vehicle to provide quantifiable benefit to the utility's business. But to do this, the SCADA planner must look beyond the traditional roles of SCADA. The planner must consider aspects of his/her project that also address the following issues:

- Reduction of substation design and construction costs
- Reduction of substation operating costs
- Overall lowering of power system operating costs
- Development of information for non-SCADA functions
- Utilization of existing resources and company standards for hardware, software, and database generation
- Expansion of automated operations at the subtransmission and distribution levels
- Improved customer relations.

To accomplish this, the SCADA planner must join forces with the substation engineer to become an integrated team. Each must ask the other "How can your requirements be met in a manner that provides positive benefits to my business?" In utilities where these questions are being asked, SCADA and substation engineers must coordinate efforts to a greater degree than ever before.

The need to reduce operating costs through substation automation is driving the replacement of existing RTUs with substation controllers. In new construction, substation controllers will become design standards to reduce both capital expenditure and future operational costs. The SCADA planner can benefit from these trends by designing a system that capitalizes on the greater capabilities of the substation controller. In turn, the SCADA planner can promote the proliferation of substation controllers; they can incorporate features in the SCADA system design that increase the return on investment for every substation controller installed.

The advanced SCADA systems of tomorrow, built upon substation controllers and LAN and WAN communications facilities, can provide the tools to generate positive, quantifiable benefits to utility construction and operation. The SCADA system planner who incorporates these tools into a SCADA strategy will become an active, contributing member in the economic posturing of his/her company for the future, a future that presents both *challenge* and *promise*. Those who fail to use their SCADA strategy to position themselves for the future will find the *challenge* beyond their solution, and the *promise* in the hands of their competitors.

The SCADA planner must look beyond the traditional roles of supervisory control and data acquisition

Acknowledgments

This is the last in a series of articles based on the IEEE *Fundamentals of Supervisory Systems* tutorial course. The tutorial includes the following chapters: Introduction and Rational for Supervisory Systems, Considerations in Applying Supervisory Control Systems to Electric Utilities, Remote Terminal Units, Telecommunications Options, Master Station Architecture, User Interface, and Project Management, and Advanced SCADA Concepts.

The tutorial was presented at the 1991 IEEE PES Transmission and Distribution Exposition and Conference (with video-taping by the IEEE Education Activities Department) and the 1992 IEEE PES Winter and Summer Meetings. An updated tutorial was presented again at the 1994 IEEE PES Transmission and Distribution Exposition and Conference.

To order the tutorial document (91 EH 0337-6 PWR) or the videotape (HV0245-1-POT, which includes one copy of the document), contact IEEE Customer Service, (800) 678-IEEE.

For Further Reading

W.E. Chainey, W.R. Block, "Recent Advances in Master Station Architecture," *IEEE Computer Applications in Power*, April 1994.

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Biographies

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Wayne R. Block is the supervisor of the Power System Computer Applications Group at the Los Angeles Department of Water and Power. He has 25 years of experience in the design of energy management systems, SCADA systems, and RTU installations. He is a Senior Member of the IEEE and the PES, a member of the Substations Committee and its Data Acquisition, Processing, and Control Systems Subcommittee, and chair of the working group on Applications of New Technologies in Substation Monitoring and Control.